

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

COURSE NAME: Artificial Intelligence

COURSE CODE:CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.(BL3)

Assessment Tool Weightage: 10%

QUESTIONS:2

Duration : 45 Minutes
Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.
2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Max Marks
Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)		
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions	
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation	
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis	
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers	
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps	
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand	

⇒ Types of Intelligent Agent:

- 1).
- Simple Reflex Agent
 - Model Based Reflex Agent
 - Goal Based Agent
 - Utility - Based Agent.

A) Simple Reflex Agent:

- * It takes decisions based only on the current percept.
- * It does not consider past experiences.

Example:

- ⇒ Turns on lights when motion is detected.
- ⇒ Switches off lights when no motion is detected.

B) Model Based Reflex Agent:

- * It maintains an internal model of the environment.
- * It remembers previous states and uses them for decision making.

Example:

- ⇒ Knows whether a room was occupied
- ⇒ keeps track of indoor temperature.
- ⇒ understands whether doors/windows are open or closed.

c) Goal Based Agent :-

- * It selects actions that help to achieve specific goal.
- * It evaluates different actions and chooses that one that best achieves the goal.

Example:

- ⇒ Maintain room temperature at 24°C
- ⇒ keep the house secure.
- ⇒ Minimize electricity usage.

d) Utility - Based Agent :-

- * It chooses actions that maximize overall benefit (utility).
- * It balances multiple objectives like comfort, energy efficiency and security.

Example:

⇒ Adjusts air conditions to save electricity while maintaining comfort.

Comparison:

Agent type	Memory	Goal oriented	learns	Suitable for smart home.
Simple	No	No	No	Basic Automation
Model Based	Yes	No	Limited	Tracks Environment
Goal Based	Yes	Yes	Limited	Achieves Objectives
Utility Based	Yes	Yes	Yes	Best overall Performance.

Best Approach:

* The most effective solution is a hybrid Agent combining.

Model-Based Agent - to remember environmental condition.

Goal Based Agent - to achieve user-defined objectives.

Utility Based Agent - to balance comfort, security and energy efficiency.

Justification :

- ⇒ Improves users comfort.
- ⇒ Reduces electricity consumption.
- ⇒ Enhances home security.

Conclusion :

A hybrid Model-Based + Goal Based Utility Based Agent is the best choice for a smart home automation system.

2) Uniformed searched strategies :

- a) uniform cost search (UCS)
- b) Iterative Deepening search (IDS).

a) uniform cost search [UCS]

- * UCS expands the node with the lowest Cumulative Path cost ($g(n)$).
- * It continues exploring the least cost path until the goal [exit] is reached.
- * Suitable when step costs are not equal.

Advantage :

- * Complete
- * Optimal - finds the least-cost path.
- * works well with varying Path costs.

Disadvantage :

- * High memory requirement.
- * Time consuming for large search spaces.

Complexity :

where b = branching factor

Time : $O(b^{1 + \frac{C^*}{\epsilon}})$ (C^* = optimal solution cost)

Space : $O(b^{(1 + \frac{C^*}{\epsilon})})$ (ϵ = minimum step cost)

b) Iterative Deepening Search [IDS]:

* IDS Performs depth limited search repeatedly with increasing depth limit until the goal is found.

* It combines the benefits of DFS and BFS.

Advantages:

* complete

* low memory usage

* Suitable when solution depth is unknown.

Disadvantages:

* Repeated search causes extra time.

* Not optimal when step costs vary.

Complexity:

Time: $O(b^d)$ where, b = branching factor

Space: $O(bd)$ d = depth of the solution.

Features	UCS	IDS.
complete	Yes	Yes
optimal	Yes	Only if step costs are equal
Time complexity	High	Moderate
Space complexity	High	Low
Best for.	Varying Path costs	Unknown depth

Relation to Intelligent Agent :

Simple reflex Agent : reacts only to current state, may get stuck in loops or dead ends.

State Based Agent : maintains state information. It avoids repeating states and builds a map.

Goal Based Agent : uses search strategies to plan actions to reach the goal (exit).

Utility - Based Agent : chooses actions that maximize utility (eg: time, safety).

Best combination :

A Model - Based and Goal - Based Agent
using uniform cost search (UCS)

Justification :

⇒ UCS guarantees the least-cost (shortest) path when movement costs vary.

⇒ This combination ensures efficient, reliable and optimal navigation in an unknown maze.